LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 6 Test (GSEB pattern): Control and Coordination

Chapter / Topic: Control and Coordination

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SECTION A

Q1. Which of the following is a plant hormone? **[1]**

Answer: (b) Cytokinin

Q2. Which part of the brain controls voluntary actions? **[1]**

Answer: (a) Cerebrum (forebrain)

Q3. The gap between two neurons is called a: **[1]**

Answer: (c) synapse

Q4. Which of these is an example of chemotropism? **[1]**

Answer: (a) A pollen tube growing towards the ovule

Q5. The brain is responsible for: **[1]**

Answer: (d) all of the above

Q6. Growth hormone is secreted by the: **[1]**

Answer: (c) pituitary gland

Q7. Which hormone regulates the blood sugar level? **[1]**

Answer: (b) Insulin (secreted by the pancreas)

Q8. Reflex arcs are formed in the: **[1]**

Answer: (d) spinal cord

Q9. The bending of a plant shoot towards light is called: **[1]**

Answer: (b) phototropism

SECTION B

Q10. What is a reflex action? Give two examples. **[2]**

Answer: A sudden, automatic response to a stimulus that happens without conscious thinking. Examples (any two): pulling the hand away from a hot object, blinking when a bright light falls on the eyes, sneezing, knee-jerk.

- Definition: 1
- Two examples: ½ + ½

Q11. What is the function of receptors in our body? What could happen if they do not work properly? **[2]**

Answer: Receptors detect information from the environment, such as light, sound, smell, taste, touch and heat, and pass it on to the nervous system. If they don't work properly, the information is not detected, so the body cannot respond correctly. For example, we might not pull our hand away from a hot object and could get hurt.

- Function: 1
- Consequence with example: 1

Q12. Why is adrenaline called the 'emergency hormone'? Give two of its effects on the body. **[2]**

Answer: It is secreted by the adrenal glands directly into the blood when we are scared or in danger, and it prepares the body to respond. Effects (any two): the heart beats faster, sending more oxygen to the muscles; blood supply to the digestive system and skin is reduced; the breathing rate increases.

- Secreted in emergency situations: 1

- Two effects: ½ + ½

SECTION C

Q13. Answer the following about a neuron: **[3]**

(i) At the dendritic tip (dendrites). [1]
(ii) From the dendrite to the cell body, and then along the axon to its end. [1]
(iii) At the end of the axon, the electrical impulse sets off the release of chemicals. These cross the gap (synapse) and start a similar electrical impulse in the dendrite of the next neuron. [1]

Q14. Name the gland: **[3]**

(i) Adrenal gland [1]
(ii) Thyroid gland [1]
(iii) Pancreas (enzymes in pancreatic juice; the hormone insulin) [1]

SECTION D

Q15. A student placed a potted plant near a window for a week. The shoot bent towards the window, while the roots grew downwards. The student read that a… **[4]**

(i) Phototropism (positive). [1]
(ii) Auxin, made at the shoot tip. [1]
(iii) Auxin diffuses to the shaded side, where the cells grow longer than on the lit side, so the shoot bends towards light. Roots show positive geotropism (they grow downwards). [2]

- Shaded side cells elongate more: 1
- Positive geotropism: 1